Causal Inference & ML

TA Session 2

Data Science for Decision Making

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2026-02-13

Outline

- Non Linear Models
- Nonparametric Density Estimation
  - Histograms
  - Kernell Density Estimation
- Non Parametric Regression Models
  - binscatter
  - NW estimator
  - Local Linear Regression
- Semiparametric Regression Models
  - Robinson’s difference estimator

Non Linear Models

The Data

- **Dataset:** CRIME4 (Wooldridge)
- **Panel structure:** 90 counties, T=7 years (1981-1987)
- **Outcome variable:** Binary indicator for high crime rate

$$\text{high\_crime}_{it} = \begin{cases} 1 & \text{if crime rate} > 0.04 \\ 0 & \text{otherwise} \end{cases}$$

- **Key regressors:**
  - `lprbarr` : log probability of arrest
  - `lprbconv` : log probability of conviction
  - `lpolpc` : log police per capita
  - `ldensity` : log population density
  - `lwcon` : log construction wage

Research question: What determines whether a county experiences high crime?

Linear Probability Model with FE

Question: How would you estimate a LPM with county fixed effects?

Model:

$$\text{high\_crime}_{it} = \alpha_i + X'_{it}\beta + \varepsilon_{it}$$

where:

- α_i = county fixed effect
- β = **marginal effects** (direct interpretation)

Within transformation get rid off α_i:

1. Demeaning and get rid off α_i

   $$\text{high\_crime}_{it} - \overline{\text{high\_crime}}_i = (X_{it} - \bar{X}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

2. Run FE regression over this transformed model

   $$\widetilde{\text{high\_crime}}_{it} = \tilde{X}'_{it}\beta + \tilde{\varepsilon}_{it}$$

```
xtreg high_crime lprbarr lprbconv lpolpc ldensity lwcon, fe vce(cluster county)
```

LPM: The Problem

1. Predictions outside the [0,1] range

$$\hat{P}(\text{high\_crime} = 1) > 1 \quad \text{(25 observations)}$$

$$\hat{P}(\text{high\_crime} = 1) < 0 \quad \text{(180 observations)}$$

2. Heteroskedasticity (by construction)

$$\text{Var}(\varepsilon_{it} \mid X_{it}) = P_{it}(1 - P_{it})$$

   Depends on X_{it}

3. Constant marginal effects (unrealistic)

   A 1% increase in arrest probability reduces high crime by **9.8 percentage points** for all counties, regardless of whether the baseline probability is 5% or 80%.

4. Imposed linearity

Nonlinear Model: Logit with FE

Model specification:

$$P(\text{high\_crime} = 1 \mid X_{it}, \alpha_i) = \Lambda(X_{it}'\beta + \alpha_i)$$

where $\Lambda(z) = \frac{\exp(z)}{1+\exp(z)}$ is the **logistic CDF**

$$P(\text{high\_crime}_{it} = 1 \mid X_{it}, \alpha_i) = \frac{\exp(\alpha_i + X_{it}'\beta)}{1 + \exp(\alpha_i + X_{it}'\beta)}$$

- Probabilities are **guaranteed** to be in [0,1],
- Better fit for binary outcomes
- Non-constant marginal effects

The IPP problem

But estimating β and 90 α_i with only $T = 7$ obsertavations creates the **Incidental Parameters Problem**

- As $N \to \infty$ but T fixed $\Rightarrow \hat{\beta}$ is **inconsistent**
- Bias = $O(1/T) \approx 14\%$

Solution: Sufficient Statistic

For the logit model,

$$S_i = \sum_{t=1}^{T} \text{high\_crime}_{it}$$

is a **sufficient statistic** for α_i

Definition: S_i is sufficient for α_i if:

$$P(\text{high\_crime}_{i1}, \ldots, \text{high\_crime}_{iT} \mid S_i, X_i, \beta) \text{ does not depend on } \alpha_i$$

What this means:

Instead of maximizing:

$$L(\beta, \alpha_1, \ldots, \alpha_N \mid y, x) = \prod_{i=1}^{N} \prod_{t=1}^{T} P(y_{it} \mid \alpha_i, x_i, \beta)$$

Maximize the conditional likelihood:

$$L_c(\beta \mid y, x, S_1, \ldots, S_N) = \prod_{i=1}^{N} P(y_i \mid S_i, x_i, \beta)$$

Result:

- α_i **disappears** from the likelihood
- Estimate β **without** estimating α_i
- $\hat{\beta}$ is **consistent** even with small T

Conditional Logit

In stata conditional Logit is implemented using `xtlogit` and `or` to interpret the coefficients as odds ratios (ORs):

```
xtlogit high_crime lprbarr lprbconv lpolpc ldensity lwcon, fe or
```

Results (N=98, 14 counties):

Variable	Odds Ratio	Interpretation
lprbarr	0.0002***	↓ Reduces odds 99.98%
lprbconv	0.0002***	↓ Reduces odds 99.98%
lpolpc	13,067***	↑ Increases odds (!)
ldensity	5,369	Not significant
lwcon	0.432	Not significant

Key findings:

- **Strong deterrence:** Higher arrest/conviction probability dramatically reduces crime odds
- **Police puzzle:** Positive coefficient likely reflects **reverse causality** (crime → hire police)
- **Sample note:** Only 14 "switcher" counties used (76 dropped due to no variation)

Interpretation: OR < 1 decreases odds, OR > 1 increases odds.

Nonparametric Density Estimation

Histograms

A nonparametric method to estimate the probability density function (PDF)

$$\hat{f}(x_0) = \frac{1}{Nh} \sum_{i=1}^{N} \frac{1}{2} \mathbf{1}\left(\left|\frac{x_i - x_0}{h}\right| < 1\right)$$

Idea: Count observations in bins, normalize by sample size and bin width

The degree of smoothing of the histogram can be controlled by selecting:

1. Number of bins

```
histogram price, bin(10)  name(hist10, replace)
```

1. Bin width $2h$

```
histogram price, width(2000) density name(hist_wide, replace)
```

Kernel Density

Smooth, nonparametric estimate of the probability density function

$$\hat{f}(x_0) = \frac{1}{Nh} \sum_{i=1}^{N} K\left(\frac{x_i - x_0}{h}\right)$$

- Role of $K(\cdot)$: weighting function that determines how much each observation x_i contributes to the density estimate at the point x_0
- Role of h: smoothing parameter that controls the width of the kernel.

How to choose optimal h?

- The one that minimizes MISE, but since we do not know f that is the true density function, we can assume that it follows a Normal distribution $f \approx N(\mu, \sigma^2)$.
- Silverman's Plug-in Solution is the one used by default in Stata.

$$h^* = 1.364 \times \delta \times \min\left(s, \frac{IQR}{1.34}\right) \times N^{-1/5}$$

For Epanechnikov kernel:

$$h^* = 0.9 \times \min\left(s, \frac{IQR}{1.34}\right) \times N^{-1/5}$$

Non Parametric Regression

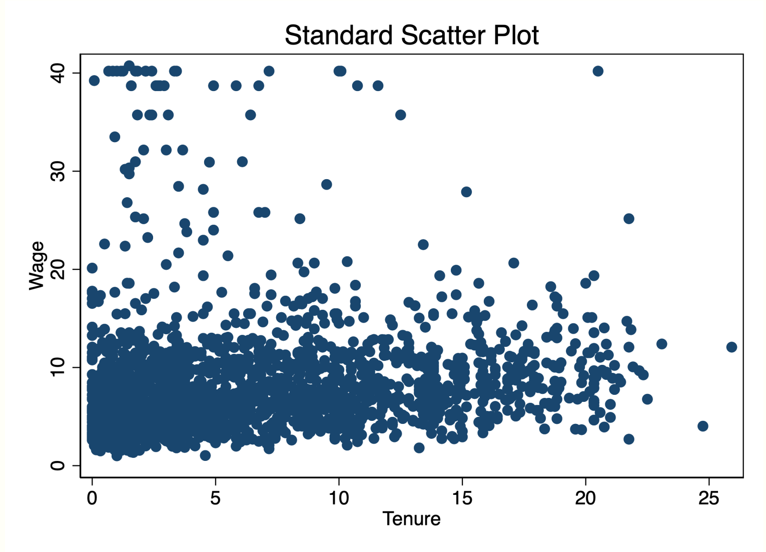
Binned Scatter plots

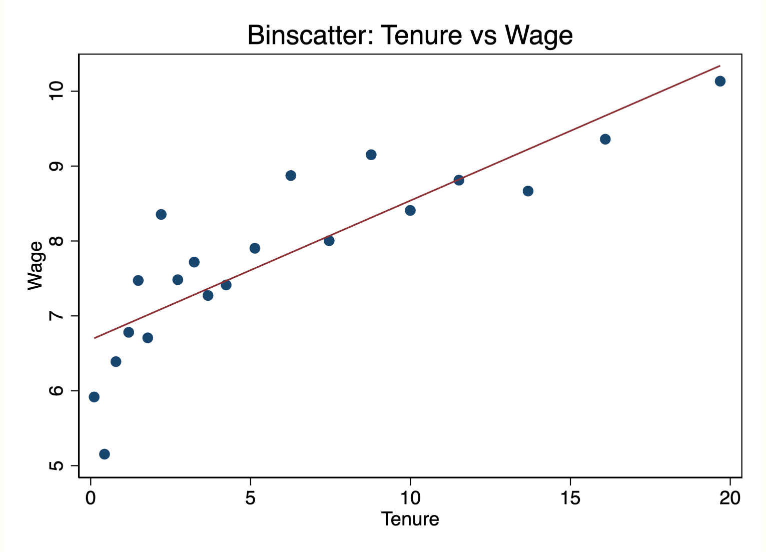
With large N, scatter plots become unreadable, and can't see conditional means:

Binscatter Steps:

1. Divide tenure into 20 bins
2. Calculate mean of wage within each bin
3. Plot (mean tenure, mean wage) for each bin

```
scatter wage tenure
binscatter wage tenure, nq(20)
```





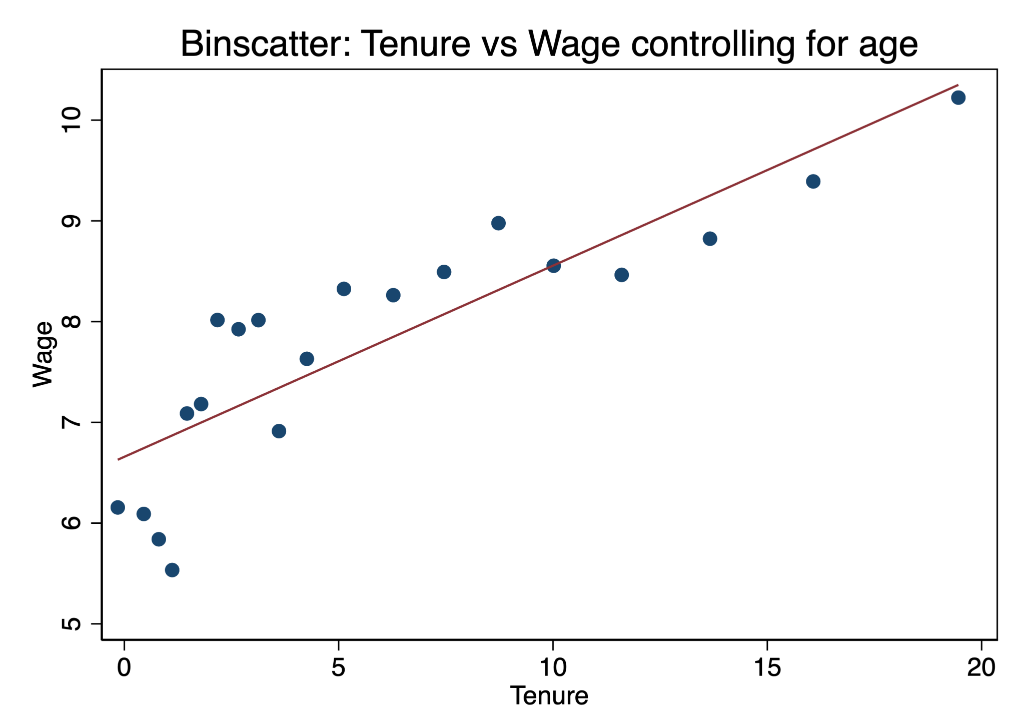
Controlling for Covariates

Frisch-Waugh-Lovell (FWL) Theorem:

1. Regress wage on controls $\longrightarrow$ get residuals $\widetilde{wage}$
2. Regress tenure on controls $\longrightarrow$ get residuals $\widetilde{tenure}$
3. Binscatter: plot $\widetilde{wage}$ vs $\widetilde{tenure}$

In Stata:

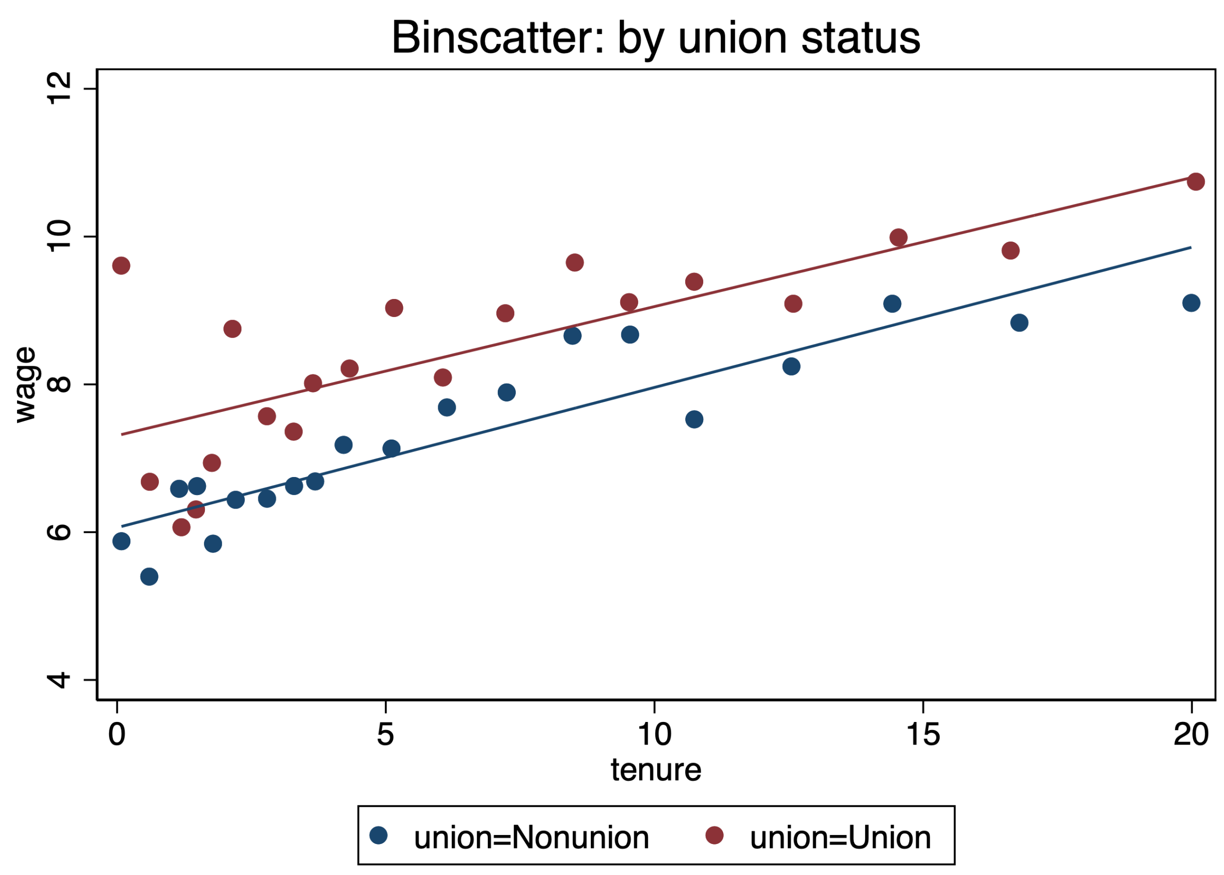
```
binscatter wage tenure, controls(age)
```



Binscatter by Groups

Compare relationships across subgroups:

```
binscatter wage tenure, nq(20) by(union)
```



Shows $E(wage|tenure)$ separately for union, useful for heterogeneity analysis.

Cattaneo et al. (2024)

Traditional binscatter issues:

1. Assummes linear CEF
2. **Bin choice matters** (10 vs 20 vs 50 bins)
3. **No formal inference** (confidence intervals)

`binsreg` **improvements:**

1. Employs semi-parametric methods to visualize CEF
2. **Data-driven bin selection** (optimal number of bins)
3. **Confidence intervals** (pointwise or uniform bands)

Nonparametric Regression

Kernel regression: Nadaraya-Watson Estimator

- `binscatter` can be seen as using an implicit rectangular (uniform) kernel.
- Kernel regression replaces this with a smooth weighting function to obtain a smooth approximation of the CEF:

$$\hat{m}_{NW}(x_0) = \frac{\sum_{i=1}^{N} K\left(\frac{x_i - x_0}{h}\right) y_i}{\sum_{i=1}^{N} K\left(\frac{x_i - x_0}{h}\right)}$$

- Points closer to x_0 get more weight
- Can be interpreted as weighted local average

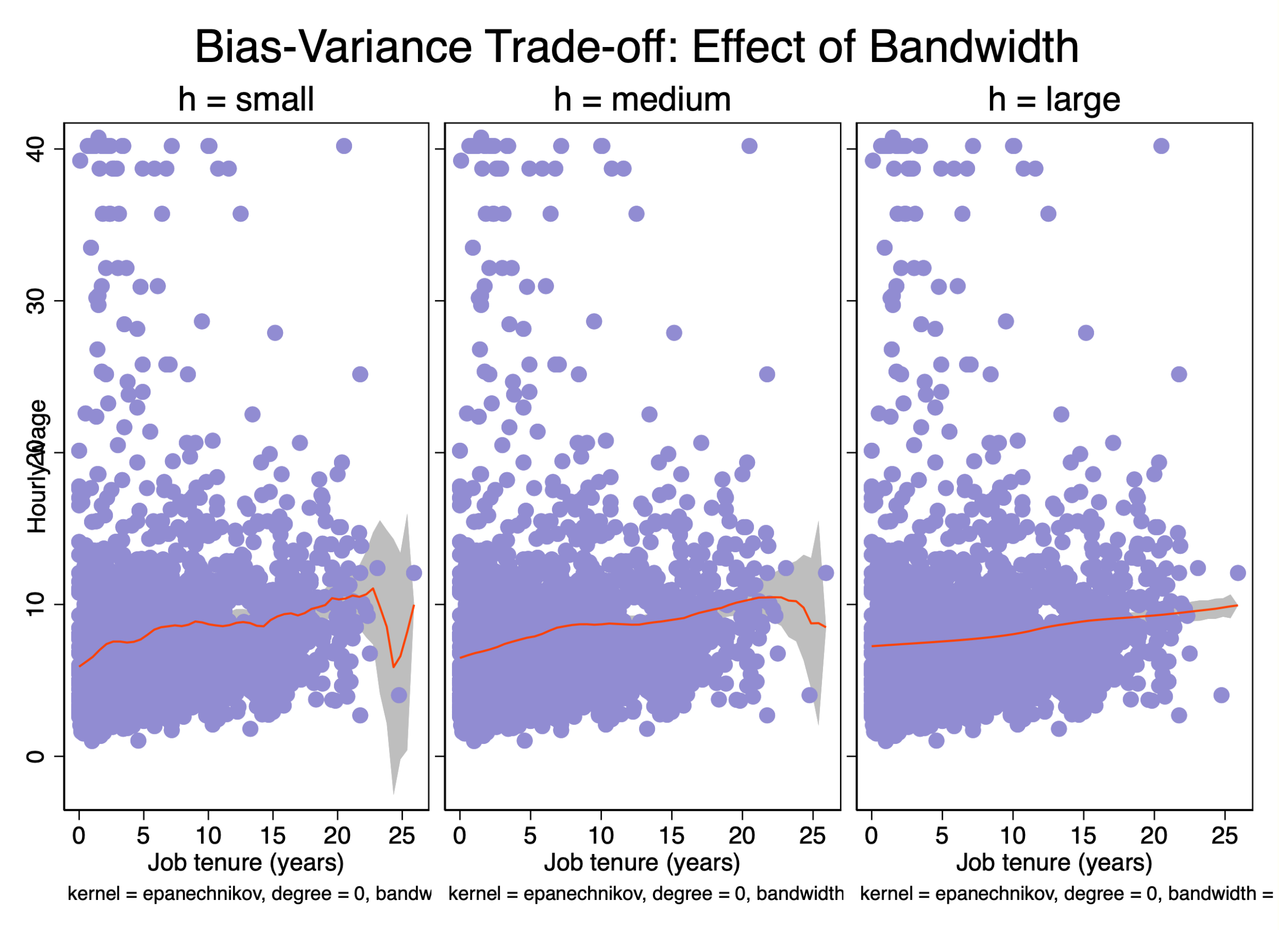
Equivalent to:

$$\min_{\beta_0} \sum (y_i - \beta_0)^2 K\left(\frac{x_i - x_0}{h}\right)$$

Bandwidth and Bias–Variance Tradeoff

As we discussed, the election of h is which control the **variance-bias** trade off.

- Small h: very local averaging → low bias, high variance (overfitting)
- Large h: heavy smoothing → high bias, low variance (underfitting)
- Intermediate h: best compromise



Choosing optimal h:

Leave-One-Out Cross Validation

$$\text{Toy data:} \quad (x, y) = \{(1,2),(2,3),(3,2),(4,5)\}$$

For a candidate bandwidth h:

1. Step 1: remove $(1, 2)$

Estimate $\hat{m}_{-1,h}(1)$ using remaining three points.

2. Step 2: prediction error:

$$(2 - \hat{m}_{-1,h}(1))^2$$

3. Step 3: Repeat for all points:

$$CV(h) = \frac{1}{4}\sum_{i=1}^{4}(y_i - \hat{m}_{-i,h}(x_i))^2$$

Choose h minimizing $CV(h)$.

In practice: `lpoly` uses plug-in bandwidth approximating this optimum.

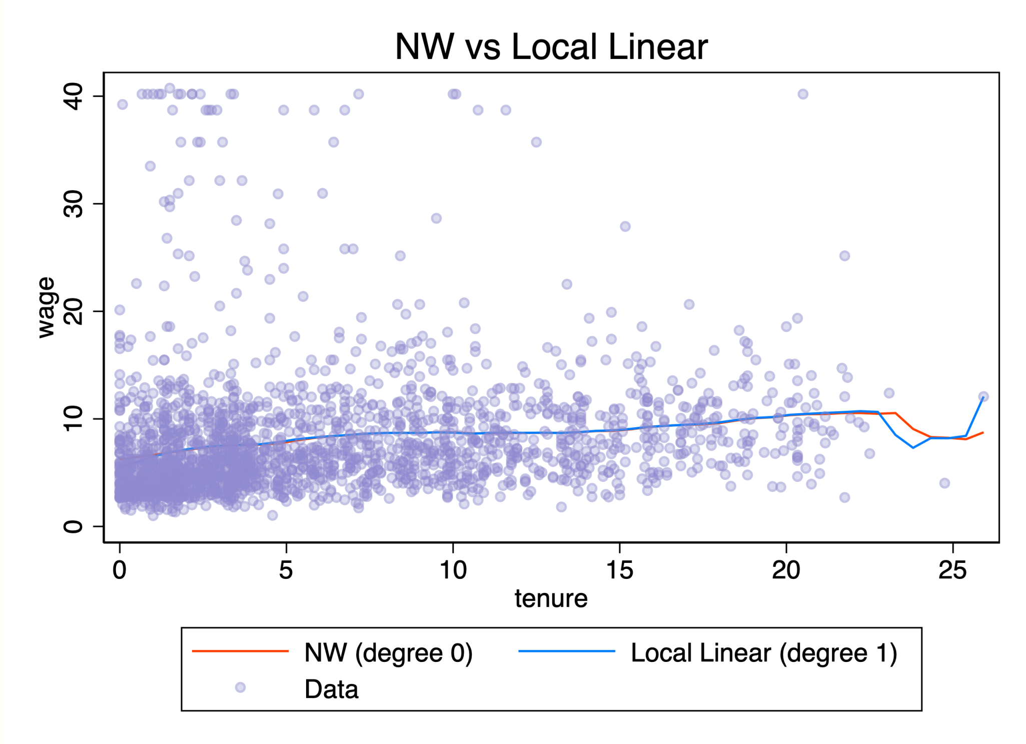
Local Linear Regression

Instead of approximating the function close to x_0 with a constant, now we can approximate it with a line.

For each x_0 solves:

$$\min_{\beta_0, \beta_1} \sum (y_i - \beta_0 - \beta_1 (x_i - x_0))^2 K\left(\frac{x_i - x_0}{h}\right)$$

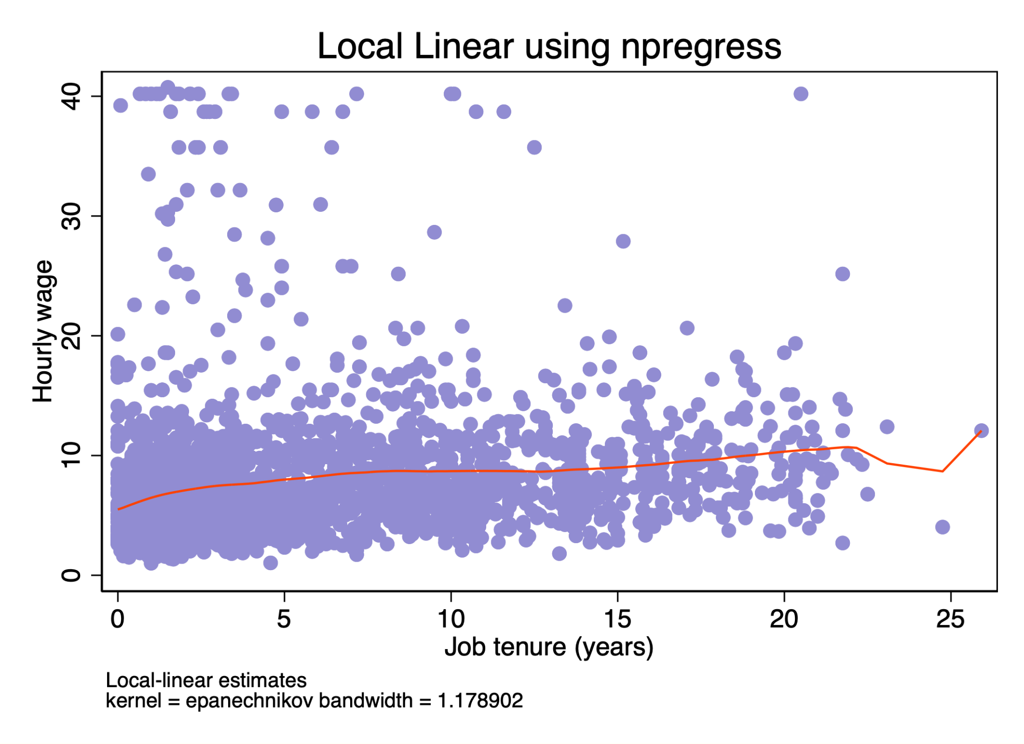
$$\hat{m}(x_0) = \hat{\beta}_0, \qquad \hat{m}'(x_0) = \hat{\beta}_1$$



npregress and Marginal Effects

```
npregress kernel wage tenure
npgraph, title("Local Linear using npregress")
```

1. Allows multiple covariates
2. Allows to make inference using (SE, VCE, robust)
3. Allows to obtain marginal effects



Semiparametric regression models

Hedonic pricing equation of houses

- **Dataset:** HPRICE3 (Wooldridge)
- **Outcome variable:** Log of house prices

$$\text{lprice}_i = X_i'\beta + \lambda(\text{ldist}_i) + u_i \quad \text{with} \quad E(u_i | X_i, \text{ldist}_i) = 0$$

- **Key regressors:**
  - `larea` : log of square footage of house
  - `lland` : log of square footage lot
  - `rooms` : # rooms in house
  - `bath` : # bathrooms
  - `age` : age of the house
  - `ldist` : log dist. from house to incin., feet

Research question: What was the effect of a local garbage incinerator on housing prices in North Andover in 1981?

House characteristics affect prices approximately **linearly**, but distance may have **nonlinear disamenity effects**.

Robinson Difference Estimator (PLM)

A nonparametric Frisch–Waugh "partialling-out" procedure.

1. Estimate conditional means (kernel):

$$\hat{E}[lprice|ldist_i] = \hat{m}_y(ldist_i) \quad \text{and} \quad \hat{E}[X|ldist_i] = \hat{m}_X(ldist_i)$$

2. Partial out distance:

$$\widetilde{lprice}_i = lprice_i - \hat{m}_y(ldist_i) \quad \text{and} \quad \tilde{X}_i = X_i - \hat{m}_X(ldist_i)$$

3. OLS:

$$\widetilde{lprice}_i = \tilde{X}_i' \beta + u_i \quad \Rightarrow \quad \hat{\beta}_{PLM}$$

4. Recover nonlinear effect:

$$\hat{\lambda}(\mathrm{ldist}_i) = \hat{m}_y(\mathrm{ldist}_i) - \hat{m}_X(\mathrm{ldist}_i)\hat{\beta}$$

Idea: remove distance nonparametrically → estimate β with clean variation → reconstruct λ.

Implementation

```
semipar lprice larea lland rooms bath age if y81==1, ///
    nonpar(ldist) robust ci ///
    title("ldist non parametric") ///
    xtitle("ldist") ///
    ytitle("{&lambda}(ldist)")
```

Parametric part:

```
                                                 Number of obs =       142
                                                 R-squared     =    0.6863
                                                 Adj R-squared =    0.6748
                                                 Root MSE      =    0.1859
------------------------------------------------------------------------------
      lprice | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       larea |   .3266051   .0887425     3.68   0.000     .1511228    .5020874
       lland |   .0790684   .0316665     2.50   0.014       .01645    .1416868
       rooms |    .026588   .0234205     1.14   0.258    -.0197244    .0729003
       baths |   .1611464   .0463598     3.48   0.001      .069473    .2528198
         age |  -.0029953    .000969    -3.09   0.002    -.0049113   -.0010793
------------------------------------------------------------------------------
```

Implementation

```
semipar lprice larea lland rooms bath age if y81==1, ///
    nonpar(ldist) robust ci ///
    title("ldist non parametric") ///
    xtitle("ldist") ///
    ytitle("{&lambda}(ldist)")
```

Non parametric part: